

20. An Algebraic Criterion for Absolute Irreducibility

Math. Ann. 85 (1922), pp. 26–33

A polynomial – with coefficients from an arbitrary, abstractly defined field – is called *absolutely irreducible* if it remains irreducible in the algebraically closed field to which the coefficient domain can be enlarged.¹ I show in what follows that absolute irreducibility, in contrast with irreducibility relative to a prescribed field, can be grasped algebraically. More precisely, the theorem is:

To every polynomial in $n \geq 2$ variables with indeterminate coefficients there can be assigned an integral rational function, with integral coefficients, of these coefficients and of further indeterminates – the reducibility form – in such a way that, for every special polynomial of the same degree, the necessary and sufficient condition for absolute irreducibility is the non-vanishing of the reducibility form. In other words: for every special system of values of the coefficients for which the reducibility form vanishes identically in the indeterminates and which does not cause a lowering of degree, the special polynomial is reducible, and conversely.

If, instead of polynomials, one considers homogeneous forms in one more variable, the degree condition is replaced by the simpler condition that the coefficient system not vanish identically.

As a direct consequence of the theorem one also obtains the following, first stated by A. Ostrowski:²

Every absolutely irreducible polynomial with algebraic numbers as coefficients remains irreducible modulo every prime ideal of a finite algebraic field Ω containing the coefficient domain, with the exception of at most finitely many prime ideals of Ω . In particular, Ω may also be the field of rational numbers.

I intend to discuss further applications to the theory of relatively integral functions³ elsewhere.

1. We first show that *every polynomial in $n \geq 2$ variables with indeterminate coefficients is absolutely irreducible*. Set

$$\begin{aligned} F(x) &= \sum A_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n} & (i_1 + \cdots + i_n \leq l), \\ G(x) &= \sum B_{j_1 \dots j_n} x_1^{j_1} \cdots x_n^{j_n} & (j_1 + \cdots + j_n \leq m), \\ H(x) &= F(x)G(x) = \sum C_{k_1 \dots k_n}(A, B) x_1^{k_1} \cdots x_n^{k_n} & (k_1 + \cdots + k_n \leq l + m), \end{aligned} \quad (1)$$

where the A and B denote indeterminates, and the $C(A, B)$ become integral bilinear combinations of the A and B . The quotients

$$D = C(A, B)/C_{0 \dots 0}(A, B)$$

are therefore rational functions of the quotients $A/A_{0 \dots 0}$ and $B/B_{0 \dots 0}$; but the number of these functions is greater than the number of their arguments, since these numbers are respectively

$$\binom{l+m+n}{n} - 1, \quad \binom{l+n}{n} - 1, \quad \binom{m+n}{n} - 1,$$

and

$$\binom{l+m+n}{n} + 1 > \binom{l+n}{n} + \binom{m+n}{n} \quad \text{for } n \geq 2, \quad l > 0, \quad m > 0.$$

Thus there is at least *one* algebraic relation among the $D(A, B)$ and consequently also among the $C(A, B)$:

$$\Phi(Z) \neq 0 \quad [\text{identically in } Z_{k_1 \dots k_n}], \quad \Phi(C(A, B)) = 0 \quad [\text{identically in } A, B], \quad (2)$$

where Φ denotes an integral polynomial.

A polynomial with indeterminate Z as coefficients,

$$E(x) = \sum Z_{k_1 \dots k_n} x_1^{k_1} \cdots x_n^{k_n} \quad (k_1 + \cdots + k_n \leq l + m), \quad (3)$$

is therefore necessarily absolutely irreducible for $n \geq 2$.⁴

¹That every field can be enlarged, essentially uniquely, to an algebraically closed one, i.e. to one in which every polynomial in one variable splits into linear factors, was shown by E. Steinitz in his *Algebraische Theorie der Körper* (J. f. M. 137 (1910), p. 167); he thereby gave the rational equivalent of the fundamental theorem of algebra.

²Zur arithmetischen Theorie der algebraischen Größen. Gött. Nachr. 1919, p. 279 (lemma p. 296). For the case of fields consisting of ordinary numbers, Ostrowski, as is known to me, also arrived at the main theorem above; he will publish his investigations on this soon. That, in my arrangement of the proof, the main theorem holds for arbitrary, abstractly defined fields rests on the fact that the reducibility form constructed for indeterminates has integral coefficients independent of the particular field taken as ground field.

³D. Hilbert, Mathematische Probleme. Gött. Nachr. 1900, p. 253, Problem 14.

⁴For, in the terminology of 2, the indeterminates Z are not zeros of $\mathfrak{P}$.

2. The totality of these polynomials $\Phi(Z)$ which vanish identically in A, B under the substitution $Z = C(A, B)$ forms an *ideal of polynomials*,⁵ more precisely a prime ideal $\mathfrak{P}$; for if, under the substitution, a product $\Phi_1\Phi_2$ vanishes, then at least one factor vanishes. Since (2) is satisfied identically in A, B , it remains valid for every special system of values A, B , where A, B and consequently also $\Gamma = C(A, B)$ denote quantities of some field. *The coefficients Γ of every polynomial reducible in an extension field therefore satisfy the conditions $\Phi(\Gamma) = 0$; they are zeros of $\mathfrak{P}$, or also of the finitely many basis polynomials of $\mathfrak{P}$.* It remains to prove the converse.

For this it should be noted that all relations among A, B, C are preserved when, by introducing a new variable y , the polynomials (1) are transformed into homogeneous forms $F(x, y), G(x, y), H(x, y)$ of dimensions $l, m, l + m$. Thus coefficients Γ also become zeros of $\mathfrak{P}$ when, for instance, $F(x, y)$ ⁶ becomes a power of y ; for them the passage to inhomogeneous polynomials causes a lowering of degree of $\bar{H}(x)$ and not a factorization. It will be shown that, with the exception of this lowering of degree, the converse always holds; in particular, that *for homogeneous forms the converse holds without exception, provided not all Γ vanish; in other words, every zero of $\mathfrak{P}$ is supplied by the parameter representation $\Gamma = C(A, B)$ with quantities A, B from an extension field.*⁷

I prove this by the direct construction of finitely many functions $\Phi(Z)$: by means of Kronecker's substitution

$$x_i = \xi^{d^{n-i}}$$

I assign to the polynomials (1) polynomials in *one* variable; I show that gaps occur here in the exponents, and by forming the norm of the corresponding coefficients I arrive at the functions $\Phi(Z)$ and at the desired criterion.

3. Put:⁸

$$d = l + m + 1, \quad i = i_1 d^{m-1} + i_2 d^{m-2} + \cdots + i_n, \quad j = j_1 d^{m-1} + \cdots + j_n, \quad k = k_1 d^{m-1} + \cdots + k_n. \quad (4)$$

Then, by Kronecker's substitution, the polynomials F, G, H in (1) correspond respectively to the polynomials

$$\begin{aligned} f(\xi) &= \sum A_{i_1 \dots i_n} \xi^i & (i_1 + \cdots + i_n \leq l), \\ g(\xi) &= \sum B_{j_1 \dots j_n} \xi^j & (j_1 + \cdots + j_n \leq m), \\ h(\xi) &= \sum C_{k_1 \dots k_n}(A, B) \xi^k & (k_1 + \cdots + k_n \leq l + m). \end{aligned} \quad (5)$$

This correspondence is known to be one-to-one, since for all induced exponents occurring in (5), $k = 0$ implies $k_1 = 0, \dots, k_n = 0$; and therefore $i + j = i' + j'$ also implies $i_1 + j_1 = i'_1 + j'_1; \dots; i_n + j_n = i'_n + j'_n$. Hence distinct products $\xi^i \xi^j$ can produce the same exponent ξ^k only when the corresponding products $x_1^{i_1} \cdots x_n^{i_n} x_1^{j_1} \cdots x_n^{j_n}$ also lead to the same exponent. Thus one obtains

$$h(\xi) = f(\xi)g(\xi), \quad (6)$$

and from the validity of a relation (6), in such a way that in $f(\xi)$ and $g(\xi)$ no exponents other than the induced ones occur, one obtains back again

$$H(x) = F(x)G(x). \quad (7)$$

Here *gaps actually occur in the induced exponents in $f(\xi)$ and $g(\xi)$* . The highest exponent of $f(\xi)$ is ld^{n-1} , while the second highest is $(l-1)d^{n-1} + d^{n-2}$; their difference is $d^{n-2}(d-1) > 1$, since $d = l + m + 1 \geq 3$ and $n \geq 2$; and the same holds for $g(\xi)$.

The relations (6) and (7), valid for the indeterminates A, B , remain valid for every special system of values A, B . To preserve the degrees, I homogenize (6) and (7) by means of the variables η and y . *A homogeneous form $H(x, y)$ therefore splits in an algebraic extension field if and only if, in that field, the corresponding*

⁵Such totalities are usually called modules; but in fact their characteristic property – that with Φ_1 and Φ_2 they contain also $\Phi_1 - \Phi_2$, and with Φ also $\Psi\Phi$, where Ψ is an arbitrary integral polynomial in Z – is the ideal property.

⁶Polynomials and forms which arise by replacing the indeterminates in $F, G, \dots, f, g, \dots$ by special systems of values will throughout be denoted by overlining: $\bar{F}, \bar{G}, \dots, \bar{f}, \bar{g}, \dots$

⁷That “in general” the zeros of $\mathfrak{P}$ are supplied by the parameter representation follows from the property of the prime ideal $\mathfrak{P}$ of defining an irreducible algebraic variety. But, as is well known, this does not imply – something I had originally overlooked and to which A. Ostrowski called my attention some time ago – that the parameter representation cannot fail for any special system of values. The proof given here rests on more elementary notions.

⁸Festschrift, p. 11.

binary form $h(\xi, \eta)$ can be split into two factors in such a way that no exponents other than the induced ones occur in the factors.

4. In order to carry out the formation of the norm mentioned at the end of 2, let

$$a_1, b_1; \dots; a_p, b_p; \quad a_{p+1}, b_{p+1}; \dots; a_{p+q}, b_{p+q}$$

be new indeterminates. Set

$$\begin{aligned} \varphi(\xi, \eta) &= (a_1\xi + b_1\eta) \cdots (a_p\xi + b_p\eta) = r_p\xi^p + r_{p-1}\xi^{p-1}\eta + \cdots + r_0\eta^p, \\ \psi(\xi, \eta) &= (a_{p+1}\xi + b_{p+1}\eta) \cdots (a_{p+q}\xi + b_{p+q}\eta) = s_q\xi^q + s_{q-1}\xi^{q-1}\eta + \cdots + s_0\eta^q, \\ \chi(\xi, \eta) &= \varphi(\xi, \eta)\psi(\xi, \eta) = t_{p+q}\xi^{p+q} + t_{p+q-1}\xi^{p+q-1}\eta + \cdots + t_0\eta^{p+q}, \end{aligned} \quad (8)$$

so that r, s, t denote the homogenized elementary symmetric functions; that is, the symmetric functions of the rows a, b that are linear in each individual row and from which all integral symmetric functions that are homogeneous and of the same dimension in each individual row are built up integrally and with integral coefficients – and in only one way.

Now form, with further indeterminates $u_{\mu\nu}$, the linear form

$$\zeta(u, a, b) = \sum u_{\mu\nu} r_\mu s_\nu, \quad (9)$$

where the summation is extended over prescribed indices μ and ν . The product of $\zeta(u)$ with all its conjugates arising from permutation of the row a, b which are different from $\zeta(u)$ and from one another – the norm $N(\zeta(u))$ of $\zeta(u)$ with respect to the field of the $t(a, b)$, when $\zeta(u)$ is regarded as a function of the field of the $r_\mu(a, b)$ and $s_\nu(a, b)$ – is then an integral symmetric function of all rows a, b , homogeneous and of the same dimension in each row. Hence, by what was said above, it is a uniquely determined integral function, with integral coefficients, of the $t(a, b)$ and $u_{\mu\nu}$:

$$N(\zeta(u, a, b)) = T(t(a, b), u) \quad [\text{identically in } a, b, u]. \quad (10)$$

The same reasoning shows that also

$$N(z - \zeta(u)) = z^\delta + T_{\delta-1}(t, u)z^{\delta-1} + \cdots + T(t, u) \quad [\text{identically in } a, b, u, z]$$

where all T are integral functions, with integral coefficients, of t and u . Both sides of this identity vanish for $z = \zeta(u)$; and indeed, because of the algebraic independence of the elementary symmetric functions $r(a, b)$ and $s(a, b)$, they vanish identically in r, s when $t(a, b)$ is set, according to (8), equal to an integral bilinear combination $w(r, s)$ of the r and s , and $\zeta(u)$ is regarded as a function of r and s . Thus

$$\zeta(u)^\delta + T_{\delta-1}(w(r, s), u)\zeta(u)^{\delta-1} + \cdots + T(w(r, s), u) = 0 \quad [\text{identically in } r, s, u]. \quad (11)$$

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The identities (10) and (11) remain valid for all special systems of values α, β of a, b , respectively ϱ, σ of r, s . If ϱ, σ are chosen in particular so that all products $\varrho_\mu \sigma_\nu$ vanish – where μ, ν denote the indices occurring in (9) – so that $\zeta(u)$ vanishes under the substitution, then, putting $w(\varrho, \sigma) = \tau$, (11) gives

$$T(\tau, u) = 0 \quad [\text{identically in } u]. \quad (12)$$

Conversely, let $\tau_0, \dots, \tau_{p+q}$ be systems of values, not all zero, for which (12) is satisfied. Since in an algebraic extension field there is a decomposition

$$\bar{\chi}(\xi, \eta) = \tau_{p+q}\xi^{p+q} + \cdots + \tau_0\eta^{p+q} = (\alpha_1\xi + \beta_1\eta) \cdots (\alpha_{p+q}\xi + \beta_{p+q}\eta), \quad (12')$$

in which α and β do not vanish simultaneously in any factor,¹⁰ although some α or β may vanish, we have $\tau = t(\alpha, \beta)$. Substituting these values into (10), using (12), shows that $\zeta(u, \alpha, \beta)$ or one of its conjugate factors vanishes identically in u . Thus, after a suitable numbering of α, β , setting $r(\alpha, \beta) = \varrho$ and $s(\alpha, \beta) = \sigma$ says precisely that $\bar{\chi}(\xi, \eta)$ admits at least one decomposition

$$\bar{\chi}(\xi, \eta) = \bar{\varphi}(\xi, \eta)\bar{\psi}(\xi, \eta) = \sum \varrho_\mu \xi^\mu \eta^{p-\mu} \cdot \sum \sigma_\nu \xi^\nu \eta^{q-\nu}, \quad (13)$$

⁹If the summation in (9) is extended over all indices, (11) represents Kronecker's extension of Gauss's theorem, essentially in Kronecker's arrangement of the proof (Zur Theorie der Formen höherer Stufen, Berliner Ber. 1883, Werke, 2, p. 417).

¹⁰This rational "fundamental theorem of algebra" is the existence theorem on which the exceptionless validity of the converse stated at the end of 2 rests.

in such a way that all products $\varrho_\mu \sigma_\nu$ occurring in $\zeta(u)$ vanish, but not all ϱ or all σ vanish.

5. The degrees p, q in (8) are now taken to be ld^{n-1} and md^{n-1} , i.e. the degrees of $f(\xi)$ and $g(\xi)$ in (5). The indices μ, ν are split into $\mu^{(1)}, \nu^{(1)}, \mu^{(2)}, \nu^{(2)}$: here $\mu^{(1)}$ runs through all exponents missing from $f(\xi)$, $\nu^{(1)}$ through all exponents of ξ in $\psi(\xi, \eta)$; correspondingly, $\nu^{(2)}$ runs through all exponents missing from $g(\xi)$, and $\mu^{(2)}$ through all exponents of ξ in $\varphi(\xi, \eta)$. Further, let

$$e(\xi) = \sum Z_{k_1 \dots k_n} \xi^k \quad (k_1 + \dots + k_n \leq l + m)$$

be the polynomial in one variable corresponding to the polynomial (3), and let

$$S(Z, u)$$

be the integral polynomial in Z and u obtained from $T(t, u)$ – the uniquely determined right hand side of (10), where the summation in (9) is extended over the indicated indices $\mu^{(1)}, \nu^{(1)}, \mu^{(2)}, \nu^{(2)}$ – by replacing the t by the coefficients Z and the corresponding zeros of $e(\xi)$.

If now r, s are replaced by the coefficients A, B and the corresponding zeros of $f(\xi)$ and $g(\xi)$, then by this substitution $\zeta(u)$ vanishes for the indicated summation. Moreover $t = w(r, s)$ passes over into the coefficients $C(A, B)$ and the corresponding zeros of $h(\xi)$; hence $T(t, u)$ passes over into $S(C(A, B), u)$. By the identity (11) we therefore get

$$S(C(A, B), u) = 0 \quad [\text{identically in } A, B, u]. \quad (14)$$

Since (14) is preserved for every special system of values, the coefficients $\Gamma = C(A, B)$ of such special $H(x)$, or more generally $H(x, y)$, that split in an extension field into two factors satisfy the condition

$$S(\Gamma, u) = 0 \quad [\text{identically in } u]. \quad (15)$$

Conversely, suppose that for the coefficients Γ , not all zero, of a polynomial $\bar{H}(x)$, respectively form $\bar{H}(x, y)$, the condition (15) is satisfied. By the above this is equivalent to $T(\tau, u) = 0$, where τ denotes all coefficients of $\bar{h}(\xi, \eta)$. Hence by (13) there exists in an extension field of the Γ a decomposition

$$\bar{h}(\xi, \eta) = \bar{f}(\xi, \eta) \cdot \bar{g}(\xi, \eta),$$

where, because of the prescribed summation, no non-induced exponents occur in $\bar{f}$ and $\bar{g}$; consequently there is a relation

$$\bar{H}(x, y) = \bar{F}(x, y) \cdot \bar{G}(x, y).$$

If, for $y = 1$, no factor becomes of degree zero – in particular if the Γ do not cause a lowering of the degree of $H(x)$ – it follows further that

$$\bar{H}(x) = \bar{F}(x) \cdot \bar{G}(x).$$

Condition (15) is therefore necessary and sufficient for $\bar{H}(x, y)$, and, when degree lowering is excluded, also for $\bar{H}(x)$, to split into two factors in an extension field.

It follows moreover that $S(Z, u)$ cannot vanish identically in Z and u , since in 1 it was shown that for $n \geq 2$ polynomials with indeterminates as coefficients, hence also the corresponding homogeneous forms in one more variable, are absolutely irreducible. Thus, with U denoting power products of the u ,

$$S(Z, u) = \sum \Phi_\lambda(Z) U_\lambda,$$

where the $\Phi_\lambda(Z)$ are, by (14), polynomials belonging to $\mathfrak{P}$. It is thereby shown that $\mathfrak{P}$ has no further zeros beyond those given by the parameter representation $\Gamma = C(A, B)$, where A and B belong to an algebraic extension field of the Γ . The converse stated at the end of 2 is therefore proved.

6. *The condition of absolute irreducibility* can now be formulated directly. One need only decompose the degree of $E(x)$ in (3), in all possible ways, into two positive summands $l + m$, and form the form $S(Z, u)$ for each such decomposition. The product over these forms,

$$R(Z, u) = \prod S(Z, u), \quad (16)$$

then forms the reducibility form of $E(x)$; for by the above, every factorization of $E(x)$, respectively $E(x, y)$, corresponds to the vanishing of one factor of (16), and conversely. Thus the main theorem stated in the introduction is proved, and at the same time it is shown how the reducibility form can actually be constructed in finitely many steps.

7. From the existence of the reducibility form (16) the theorem of Ostrowski mentioned in the introduction follows immediately. Let

$$\bar{E}(x) = \sum \Gamma_{k_1 \dots k_n} x_1^{k_1} \cdots x_n^{k_n} \quad (k_1 + \cdots + k_n \leq v)$$

be an absolutely irreducible polynomial with algebraic integers Γ as coefficients; let Ω denote a finite algebraic number field containing all Γ , and let $\mathfrak{p}$ be a prime ideal of Ω .

Then the reducibility form $R(Z, u)$ of exact degree for $\bar{E}(x)$ remains different from zero under the substitution $Z = \Gamma$; that is,

$$R(\Gamma, u) = \sum P_\lambda U_\lambda \neq 0 \quad [\text{identically in } u].$$

The ideal $\mathfrak{t} = (P_1, P_2, \dots)$ is therefore divisible by only finitely many prime ideals of Ω . If $\mathfrak{p}$ is different from these finitely many, then $R(\Gamma, u)$ does not vanish identically modulo $\mathfrak{p}$; and since the residue classes modulo $\mathfrak{p}$ again form a field, it follows that $\bar{E}(x)$ modulo $\mathfrak{p}$ is irreducible, more precisely absolutely irreducible. The same holds for $\bar{E}(x, y)$, so that no lowering of degree modulo $\mathfrak{p}$ occurs.

If the coefficients of $\bar{E}(x)$ are algebraically fractional numbers, then $\mathfrak{p}$ must also be taken different from the finitely many prime ideals dividing the common denominator Λ ; modulo all remaining prime ideals, $E(x)$ may be replaced by $\Lambda \bar{E}(x)$, so that the preceding hypotheses are satisfied. This proves the theorem.

Erlangen, August 1921.

(Received 28. 8. 1921.)

21. Formal Variational Calculus and Differential Invariants

Encyclopedia of the Mathematical Sciences II, 3 (1922), pp. 68–71 (in: R. Weitzenböck, Differential Invariants)

28. Formal Variational Calculus and Differential Invariants.¹⁴⁹

The generation of all differential invariants and the derivation of the main theorems can be carried out most uniformly by the formal methods of the calculus of variations. This principle goes back to Riemann⁷²⁾ and received its chief form-theoretic development from Lipschitz⁷³⁾.

Start from the variational problem belonging to a homogeneous form $f(dx)$ – where

$$\left| \frac{\partial^2 f}{\partial dx_i \partial dx_k} \right| \neq 0$$

is assumed – namely

$$\delta \int_{t_0}^{t_1} f(x') dt = 0, \quad \left(x'_i = \frac{dx_i}{dt} \right). \quad (140)$$

By partial integration one is led to the invariant system of equations

$$2\psi_i = \frac{d}{dt} \frac{\partial f}{\partial x'_i} - \frac{\partial f}{\partial x_i} = 0,$$

where the Lagrange expressions ψ_i are contragredient to the dx . Formally this is seen most simply if the ψ_i are defined by the integral-free formal identity corresponding to partial integration (Lagrange central equation¹⁵⁰)

$$\delta f - df_\delta = -2 \sum \psi_i(d, d) \delta x_i, \quad \left(f_\delta = \sum \frac{\partial f}{\partial dx_i} \delta x_i \right),$$

where both f and df_δ are invariants, and hence the right hand side is one as well. In particular, for a quadratic form one obtains

$$\delta \sum g_{ik} dx_i dx_k - 2d \sum g_{ik} dx_i \delta x_k = -2 \sum \psi_\mu(d, d) \delta x_\mu.$$

Replacing d here by the cogredient $(d + \lambda \delta)$ and comparing powers of λ , one obtains the corresponding system of identities:

$$\left\{ \begin{array}{l} D \sum g_{ik} dx_i dx_k - 2d \sum g_{ik} dx_i D x_k = -2 \sum \psi_\mu(d, d) D x_\mu, \\ D \sum g_{ik} dx_i \delta x_k - \delta \sum g_{ik} dx_i D x_k - d \sum g_{ik} \delta x_i D x_k = -2 \sum \psi_\mu(d, \delta) D x_\mu, \\ D \sum g_{ik} \delta x_i \delta x_k - 2\delta \sum g_{ik} \delta x_i D x_k = -2 \sum \psi_\mu(\delta, \delta) D x_\mu. \end{array} \right. \quad (141)$$

It follows that $\psi_\mu(d, \delta)$, and in particular also $\psi_\mu(d, d)$ and $\psi_\mu(\delta, \delta)$, are contragredient vectors. From (141) one computes $\psi_\mu(d, \delta)$ as

$$\psi_\mu(d, \delta) = \sum g_{i\mu} d \delta x_i + \sum \left[\begin{smallmatrix} ik \\ \mu \end{smallmatrix} \right] dx_i \delta x_k,$$

and from this one obtains the cogredient vector

$$p^\sigma(d, \delta) = \sum g^{\sigma\mu} \psi_\mu = d \delta x_\sigma + \sum \left\{ \begin{smallmatrix} ik \\ \sigma \end{smallmatrix} \right\} dx_i \delta x_k. \quad (142)$$

Putting $p^\nu(d, d) = 0$ evidently gives the equations of the geodesic lines.

Knowledge of the cogredient vector $p^\sigma(d, \delta)$ leads directly to the covariant derivative (compare No. 19). If, for instance, $h(d, \delta)$ is a form of the two rows dx and δx , then the expression

$$h^{(1)}(dx, \delta x) = \delta h - \sum \frac{\partial h}{\partial dx_\sigma} p^\sigma(d, \delta) - \sum \frac{\partial h}{\partial \delta x_\sigma} p^\sigma(\delta, \delta) \quad (143)$$

is itself an invariant as a difference of invariants, and is formed so that the second differentials cancel. Thus $h^{(1)}(dx, dx)$ is a form involving only first differentials: the covariant derivative of h . The corresponding statement holds when h contains more rows $d^{(1)}x, \dots, d^{(\lambda)}x$.¹⁵¹

¹⁴⁹This number is due to E. Noether.

¹⁵⁰For special f the designation is due to Heun; see his article IV, 1 11, p. 447.

¹⁵¹Lipschitz, J. f. Math. 72 (1870), p. 17.

The formation of the curvature form by Riemann and Lipschitz rests on the same principle. One passes from $\delta^2 f$ to the “normal form of the second variation”

$$\Omega = \delta^2 \sum g_{ik} dx_i dx_k - 2d\delta \sum g_{ik} dx_i \delta x_k + d^2 \sum g_{ik} \delta x_i \delta x_k, \quad (144)$$

which, as an aggregate of invariants, is itself again an invariant, and in which, now generalizing the Lagrange central equation, the third differentials are eliminated. The elimination of the second differentials is carried out, in analogy with covariant differentiation, by forming the difference

$$K = \Omega - 2 \sum \{p^\sigma(d, \delta) \psi_\sigma(d, \delta) - p^\sigma(\delta, \delta) \psi_\sigma(d, d)\}, \quad (145)$$

which may also be written as¹⁵²

$$\begin{aligned} \frac{1}{2}K &= d \sum \psi_\sigma(d, \delta) \delta x_\sigma - \delta \sum \psi_\sigma(d, d) \delta x_\sigma \\ &\quad - \sum \{p^\sigma(d, \delta) \psi_\sigma(d, \delta) - p^\sigma(\delta, \delta) \psi_\sigma(d, d)\}. \end{aligned} \quad (146)$$

The law of formation of K may also be stated by saying that in Ω the second differentials are eliminated by setting $p(d, \delta)$ and $p(\delta, \delta)$ – respectively the right hand sides of (141) – equal to zero. This is Riemann’s definition of the curvature form (compare No. 21), but it must be explicitly noted – as (146) makes especially clear – that this setting to zero may take place only after the differentiation has been performed; Lipschitz’s formal mode of formation avoids this difficulty. Correspondingly, the covariant derivative $h^{(1)}$ (143) could also be defined by eliminating in δh the second differentials by setting $p(d, \delta), \dots$ equal to zero.

From the curvature form one obtains, by successive formation of covariant derivatives, a sequence of forms $[\Phi_4, \Phi_5, \dots$ in Christoffel], whose projective invariants, after adjoining f , exhaust according to Christoffel and Ricci the totality of the differential invariants of the quadratic form, and whose equivalence under linear transformations, without any further integrability condition, expresses the equivalence of two quadratic differential forms (compare No. 22). The inner reason for this reduction theorem rests on the existence of the Riemannian normal coordinates arising from the variational problem (140) (compare No. 20): these transform the geodesic lines through a point into straight lines, and consequently transform linearly under arbitrary transformations of the x . The formation of all invariants and the equivalence problem are thereby reduced to the corresponding question for the individual homogeneous components in the expansion of f in normal coordinates, and one shows that these components are linearly composed from $f, \Phi_4, \Phi_5, \dots$. For quadratic forms this is carried out in detail by Vermeil⁹²), following a note by E. Noether⁹³). According to that note, the theorems and methods remain valid when an arbitrary differential expression is taken as the basis (compare Nos. 22 and 23); the scope of the methods of formal variational calculus as against those of pure elimination theory is thereby made apparent.¹⁵³

Levi-Civita¹³⁴) interpreted the setting $p(d, \delta) = 0$ geometrically as “parallel displacement of a vector” (compare also Hessenberg¹³³). This concept, cast axiomatically by Weyl, is essentially restricted to quadratic forms, but it admits a generalization of another kind. It remains valid when the group is enlarged (multiplication of the g_{ik} by an arbitrary function), as soon as, besides the quadratic form, a linear form is also made basic. Then $p(d, \delta)$ is uniquely determined, invariant constructions corresponding to those above can be carried out, and geodesic coordinates can be defined; but $p(d, d) = 0$ now no longer arises from a variational problem.¹⁵⁴ Questions about the reduction theorem and equivalence have not yet been attacked here in general; only for invariants of lowest order has R. Weitzenböck given a reduction theorem.¹⁵⁵

Finally, attention should be drawn to the general “invariant variational problems”, i.e. to those in which the simple or multiple integral J is invariant with respect to some group in Lie’s sense. Special physically interesting cases are treated in works by Hamel¹⁵⁶, Herglotz¹⁵⁷, Lorentz¹⁵⁸, Fokker¹⁵⁹, Weyl¹⁶⁰ and Klein¹⁶¹. The principled formulation by E. Noether¹⁶² shows that to the invariance of J under a G_ρ (a finite group

¹⁵²Lipschitz, J. f. Math. 82 (1876), § 1.

¹⁵³Compare these remarks, and for further geometric interpretations the seminar lectures of F. Klein listed under “Literature.”

¹⁵⁴H. Weyl, Reine Infinitesimalgeometrie, Math. Ztschr. 2 (1918), p. 384–411, and Raum, Zeit, Materie (3rd and 4th ed.).

¹⁵⁵Wien. Ber. 129 (1920), p. 683 and 697.

¹⁵⁶Math. Ann. 59 (1904), p. 416–434; Ztschr. Math. Phys. 50 (1904), p. 1–54.

¹⁵⁷Ann. d. Phys. (4) 36 (1911), p. 511, especially § 9.

¹⁵⁸Verslag der Amsterd. Akad., April, Sept. 1916, Oct. 1916, May 1917.

¹⁵⁹Ibid., Jan. 1917.

¹⁶⁰Ann. d. Phys. (4) 54 (1917), p. 117, § 2.

¹⁶¹Gött. Nachr. 1917, p. 469–482; *ibid.* 1918, p. 171–189.

¹⁶²Gött. Nachr. 1918, p. 235–257.

with ρ essential parameters) there correspond ρ linearly independent divergences; to the invariance under an infinite group containing ρ arbitrary functions up to the σ -th derivative there correspond ρ dependencies between the Lagrange expressions and their derivatives up to order σ . In both cases the converse holds. Since the Lagrange expressions become relative invariants of the group, one has at the same time a process generating invariants.

(Completed in March 1921.)